

Data Exploratory on Taxi Data in New York City

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Abstract—With increasing amounts of urban data being collected and made available, new possibilities of data-driven research emerge lead to changes in people living through scientific proof-based decision-making and strategies. In this paper, the researcher concentrates on an urban data set of particular significance which is taxi rides. Taxis are essential instruments, and taxi-related information can provide ground-breaking insight into different facets of city life, from economic activity and human behavior to mobility trends. Taxi data involves geographical components as well as several variables associated with each ride. Throughout this research, data insight is gain through data pre-processing, feature engineering and data exploratory analysis on the city transportation. The data exploration reveals the time, geo-location details, taxi fare and optimal fleet size of taxi transportation in the city. This transformation of raw data enables researcher, data scientist to have the meaningful insight into the data and understand the mobility pattern of New York City.

Keywords—Data exploratory, Feature Engineering, Geolocation data, Urban understanding, Taxi dataset.

I. INTRODUCTION

Taxi is one of the urban public transport in many busy countries. Unlike other public transports, taxi rides provide accessibility, convenience, yet privacy to passengers [1]. A competitive and reasonable taxi pricing is worth the ride for private car users to switch to a taxi service. Millions of taxi trips data are generated on monthly basis, which this data can be useful to gain the insight of the traffic patterns and obtain a clear view of urban city life [2]. Not only that, by leveraging the given dataset, taxi demand on major events like Christmas and New Year's Eve, can be studied in order to make a better decision making.

The burgeoning field of geographic information science is ushering in a transformative era in our understanding and planning of urban landscapes [3]. At the heart of this evolution lies the concept of spatial assembly, a critical analytical step that not only summarizes but also enhances our perception of the geographic environment, particularly in the context of the demand for taxi services [4]. The advent of automated data collection related to taxi movement opens a window into a city's operational dynamics, enabling the extraction of valuable insights from geospatial data through both spatial and temporal lenses. This nuanced approach promises a more accurate and comprehensive depiction of the intricate nature of a region, with daily movement and activities embedded in geospatial data serving as indicators of the socio-economic properties underpinning urban functions [5].

Geographic information science, as a burgeoning discipline, has emerged as a linchpin in reshaping our comprehension of urban environments and refining the methodologies for effective urban planning [3]. This scientific progression unfolds against the backdrop of a digital age where the proliferation of spatial data and advanced technologies has become a catalyst for a more holistic understanding of urban dynamics.

Spatial assembly, within the realm of geographic information science, takes center stage as a pivotal analytical process with profound implications for interpreting the demand for taxi services in urban landscapes [4]. At its core, spatial assembly involves the synthesis and aggregation of geospatial data, allowing for the consolidation of diverse elements that contribute to the complex tapestry of urban environments. This analytical step is instrumental in distilling the richness of geographic information into actionable insights that can inform urban planners, policymakers, and stakeholders. The demand for taxis, a quintessential aspect of urban mobility, becomes a focal point for the application of spatial assembly techniques. By harnessing automated data collection methods related to taxi movement, cities gain the ability to unravel intricate patterns embedded in geospatial data. This, in turn, facilitates a nuanced understanding of a city's operational dynamics, transcending traditional approaches to urban planning.

The significance of automated data collection in the context of taxi movement cannot be overstated. It represents a paradigm shift from conventional methods to a more dynamic and real-time approach. This transition allows for the extraction of a city's operational nuances from both spatial and temporal perspectives. The integration of temporal dynamics adds a layer of sophistication, enabling urban planners to decipher not only the spatial distribution of taxi demand but also the temporal ebbs and flows, reflecting the pulse of the city over time.

In practical terms, the insights derived from geospatial data related to taxi movement serve as a powerful lens through which to view the socio-economic properties underpinning urban functions [5]. Daily movements and activities, as captured in the geospatial dataset, become invaluable indicators of the dynamic interplay between urban structures and human behaviors. The spatial and temporal dimensions embedded in the data offer a holistic representation of a region's vibrancy, capturing the ephemerality of daily life and the fluidity of urban activities.

The spatial dimension of geospatial data allows for the mapping of taxi demand across different areas of a city. This spatial distribution becomes a reflection of the urban landscape, highlighting hotspots of activity, transit hubs, commercial centers, and residential clusters. The temporal dimension, on the other hand, adds a dynamic element, unraveling the temporal patterns of taxi demand throughout the day, week, or even season. These temporal insights capture the rhythm of urban life, including peak commuting hours, leisure periods, and the cyclical nature of activities.

The synthesis of spatial and temporal perspectives through spatial assembly techniques transcends mere data analysis; it transforms data into actionable intelligence for urban planners. By discerning patterns in taxi demand, planners can optimize transportation infrastructure, streamline traffic flow, and enhance the overall efficiency of urban mobility systems. The real-time nature of automated data collection ensures that

these insights remain current and relevant, providing a basis for agile and responsive urban planning.

Moreover, the social-economic properties embedded in daily movement and activities become discernible through the lens of geospatial data. The ephemerality of human behaviors, captured in the dynamic interplay of spatial and temporal elements, paints a vivid picture of the city's socio-economic landscape. Commercial districts may exhibit distinct patterns of activity during business hours, residential areas may see heightened demand during evening hours, and cultural or recreational hubs may experience surges during weekends. These nuances offer a rich tapestry of information, allowing urban planners to align infrastructure development with the diverse needs and rhythms of urban life.

The application of spatial assembly techniques to analyze taxi demand is particularly relevant in the context of smart cities, where data-driven decision-making is a cornerstone of urban governance. The fusion of automated data collection with advanced analytics not only empowers cities with a real-time understanding of their operational dynamics but also enables them to proactively respond to emerging trends and challenges. Smart urban planning, underpinned by spatial assembly methodologies, leverages the wealth of geospatial data to create responsive and resilient urban ecosystems. The symbiotic relationship between geographic information science, spatial assembly, and automated data collection in the realm of taxi demand analysis reshapes the landscape of urban understanding and planning. The fusion of spatial and temporal perspectives through spatial assembly techniques offers a nuanced and comprehensive depiction of a city's operational dynamics. The demand for taxis serves as a focal point, unraveling the intricate patterns embedded in geospatial data and providing insights into the socio-economic properties that underpin urban functions. This transformative approach not only enhances our understanding of urban landscapes but also equips urban planners with actionable intelligence to optimize infrastructure, streamline transportation, and foster the development of responsive and resilient smart cities. As technology continues to advance, the synergy between geographic information science and urban planning promises to usher in a new era of dynamic, data-driven urban development.

This paper will be focusing on examining the urban dataset of New York City taxi trips. According to taxi fact book report release by NYC Taxi & Limousine Commission (2018), TLC taxi trips in New York City has a total of over 41 million trips a year between June 2017 and June 2018 [6]. In general, taxi trips comprise spatial elements like longitude and latitude points. Thus, this data enables data scientist to encode geolocation information into an insight of urban traffic movement and activities.

Not only that, other attributes like taxi Id, total fare amount, and number of passengers are also recorded which allows researchers to study the traffic congestion, economics of fare pricing, and optimal fleet size [3]. Not surprisingly, it is challenging to explore those data and results. Commonly used tools and platforms like SAS, Python, Tableau are used in this report. Although analyzing high load of data is hard to perform, scientist first selects a smaller set of data to be analyzed as an alternative.

II. RELATED WORKS

Yanga, et al. [7] proposed a study to investigating the relationship between demand for taxi, land usage, and accessibility in Washington D.C. using GPS and GIS data. These datasets of latitude and longitude values were used to visualize geographic data to discover traffic densities. Further illustration in taxi trips was provided and other data exploration was performed including missing data handling. the model showed that residential density, employment density, and larger block size have high correlation with taxi demand. Not only that, he also revealed that demand for taxi was lower in the area nearby bus stops and the reverse was true in metro stations exist. Additionally, during winter season the demand for taxis was reduced.

The recent study by Zhu, et al. [8] leveraging three-month taxi trajectory in Beijing, China to explore the urban mobility by analyzing the spatio-temporal pattern on streets, specifically on 5th Ring Road. Before analysis process, data was cleaned, grouped and sorted. Non-correlated attributes were eliminated and merely geolocation records were included for traffic analysis which were pickup and drop off points, date, time, and lastly, the status of trajectory. Data was transformed from coordinate-based taxi trip into a street-based trip by implementing simplified map-matching process techniques. As the result, the transformation of geolocation data into street unit facilitates urban planners and government policy makers to have a better understanding of urban structures, optimizing public resources, and have a greater sense of urban movement in every streets unit.

Kong, et al. [9] proposed a taxi allocation model using three-dimensional attributes namely: time, location and relationship. In their work, the correlation between passengers' getting on and off on particular time and location was deeply explored by leveraging numerous data-driven analysis techniques on taxi trajectory datasets. The aim of Time-Location-Relationship (TLR) taxi matching model was to increase the opportunity of finding vacant taxis for passengers and at the same time reduce transportation resource consumption and air pollution.

Kong, et al. [10] experiments revealed that there was a rise of get-on demand taken place in diplomacy areas for affairs purposes during morning time and immediately leave during lunch time, which contributes a get-on and off demand in restaurants and café districts. It has been highlighted that commercial entertainment areas like shopping malls has higher get-on demand at the peak at 12 PM. Not only that, the taxi demand is lesser during weekend in railway exits area. There was no doubt that data exploration on the pattern of passengers' going up and down in particular time and location can be useful to develop a better model to allocate drivers and make more profitable income to drivers. Not only that, it also reduces the number of resources used during taxi vacant time.

To sum up, data is the biggest asset for urban city planning, governments, and institution to solve the real-world traffic problems such as optimizing transportation resources, organizing major events, reduce traffic congestion, sensing the urban mobility and many more. However, urban traffic insight is hard to analyze without data exploration process. Therefore, data preparation and feature engineering are the necessary tasks to transform raw information into meaningful insight of the traffic condition of big cities.

III. PROPOSED METHOD

In this paperwork, data is obtained through Kaggle repository titled “New York Taxi Fare Prediction” which the data composes of taxi ride in New York City from 2009 to 2015. The dataset attributes including key, pickup and drop-off geolocation, date, time, passenger count and finally fare amount will be analyzed. In total, the given dataset composes of nine core attributes.

It is the best practice to handle missing data and remove outlier before proceeding to data exploration. Missing data happens when data point is not recorded for an observation within a dataset [11]. In this case, technique like mean imputation is used to replace any missing data present in the NYC taxi dataset. Missing data in fare amount attribute and geolocation points are replaced using mean imputation. Additionally, there are outlier data present in this taxi dataset. Outlier is an observation in a random sample from a population that lies an abnormal distance from other values. A mistake during data collection might introduce the outlier in dataset or it is just an indication of variance in the dataset [12].

As taxi dataset is dealing with geolocation both latitude and longitude data. Hence, boxplot or other statistic functions cannot be used to identify outliers in geospatial data. Therefore, in order to get a better view of data, latitude and longitude should plot the coordinates on a map to get a better view. Since the scope of taxi data is focused on yellow taxis, which yellow cabs are the only vehicles which have the right to pick up prearranged passengers and anywhere in New York City. Hence, it is necessary to identify the NYC coordinates [13].

The utilization of density maps in the analysis of pickup taxi demand unveils a nuanced understanding of temporal patterns, revealing the ebb and flow of demand throughout the day. This analytical tool serves as a dynamic canvas, illustrating the spatial distribution of pickup demands based on a timely basis. Notably, the density map exhibits a distinctive dip in pickup demand during the early morning hours, specifically from 04 AM to 06 AM. This temporal window is characterized by a solitary, diminutive red point on the map, signifying a notable reduction in taxi demand during this period.

As the day progresses, the pickup demand experiences a discernible upswing, gaining momentum from 07 AM to 09 AM. This timeframe marks the commencement of heightened demand, gradually building up towards a crescendo. The peak of pickup demand materializes during the evening hours, particularly from 07 PM to 09 PM, where the density map is punctuated by an array of vibrant red spots. This visual representation underscores the concentration of taxi demand during these peak hours, emphasizing the temporal dimension of urban mobility patterns.

A noteworthy observation surfaces during the nighttime hours, specifically from 10 PM to 03 AM, as the density map indicates an increased prevalence of taxi trips originating outside Manhattan City. This spatial shift in demand suggests a dynamic nocturnal mobility pattern, potentially influenced by factors such as nightlife activities or residential locations outside the central business district.

In contrast, the dynamics of pickup demand diverge during the morning hours, experiencing a nadir in demand from 07

AM to 09 AM. However, this subdued period is transient, as the demand gradually intensifies, reaching its zenith during the late-night hours, specifically from 01 AM to 03 AM. This temporal asymmetry in pickup demand underscores the fluctuating nature of urban mobility needs, influenced by varying factors throughout the day.

An intriguing spatial dimension emerges when examining taxi trip destinations during the evening hours, particularly from 07 PM to 04 AM. This timeframe witnesses an increased volume of taxi trips traversing the Brooklyn area. The spatial distribution depicted on the density map reflects a concentration of red spots in Brooklyn, reaching its pinnacle from 01 AM to 03 AM. This spatial shift in taxi trips reveals the diversification of mobility patterns, with a significant portion of the population opting for taxi services in the Brooklyn vicinity during these nocturnal hours.

Delving into the correlation between pickup distance from the New York City center and taxi fare trip analysis unveils a compelling relationship. The analysis suggests that the distance of the pickup location from the city center is intricately linked to the fare structure of taxi trips. Specifically, a discernible trend emerges, indicating that the farther the pickup distance from the city center, the higher the associated taxi fare trips. This correlation underscores the influence of spatial parameters on fare dynamics, reflecting the intricate interplay between geographical locations and pricing structures.

The temporal nuances embedded in the density map offer a temporal tapestry of urban mobility patterns, shedding light on the dynamic interplay between time of day and taxi demand. The dip in demand during the early morning hours aligns with the typical lull in urban activities, as denoted by the solitary red point on the map. The subsequent surge in demand from 07 AM to 09 AM corresponds to the awakening of urban life, with commuters seeking taxi services for their morning journeys.

The crescendo of pickup demand during the evening peak hours, from 07 PM to 09 PM, signifies a confluence of factors such as rush hour traffic, social activities, and the transition from work to leisure. The vibrancy of the red spots on the density map during this timeframe encapsulates the heightened urban mobility dynamics, reflective of the city's pulse during its busiest hours. The spatial shift in pickup demand during the nighttime hours, particularly from 10 PM to 03 AM, introduces an intriguing narrative. The prevalence of taxi trips originating outside Manhattan City during this timeframe suggests a distinct nocturnal mobility pattern, possibly influenced by nightlife, events, or residential choices outside the city center. This spatial diversification highlights the adaptability of taxi services to cater to evolving urban needs across different temporal phases.

The temporal asymmetry in pickup demand, with a dip during the morning and a peak during the late-night hours, underscores the dynamic nature of urban mobility. This fluctuation is a testament to the evolving patterns of daily life, shaped by diverse activities, work schedules, and social dynamics. The density map, acting as a visual chronicle of these temporal dynamics, becomes a valuable tool for urban planners and policymakers seeking to optimize transportation resources and enhance the efficiency of mobility systems. The exploration of taxi trip destinations adds a spatial layer to the

narrative, revealing distinct patterns during the evening hours. The concentration of taxi trips in the Brooklyn area, especially from 07 PM to 04 AM, signifies a spatial preference during these nocturnal hours. The peak from 01 AM to 03 AM suggests a surge in demand for taxi services in Brooklyn during the late-night period, potentially driven by nightlife activities, events, or residential considerations.

Transitioning to the correlation between pickup distance and taxi fare trip analysis, a compelling relationship emerges. The analysis indicates that the spatial parameter of pickup distance from the city center plays a pivotal role in shaping the fare structure of taxi trips. The observed trend underscores the economic implications of spatial dynamics, with fares escalating as the pickup location moves farther away from the central business district. This correlation is indicative of the complex interplay between geographical considerations and fare calculations, highlighting the need for a nuanced understanding of the spatial dimensions influencing urban mobility costs. The density map, when harnessed for analyzing pickup taxi demand on a timely basis, serves as a potent tool for unraveling the intricacies of urban mobility patterns. Its visual representation provides insights into the temporal and spatial dynamics of pickup demand, offering a holistic view of urban life. The temporal patterns, from the early morning lull to the evening crescendo, depict the city's heartbeat, echoing the rhythm of daily life. The spatial nuances, including the nocturnal shift in taxi trip destinations and the correlation between pickup distance and fare analysis, contribute to a multifaceted understanding of urban mobility. As cities continue to evolve, the integration of density maps and advanced analytics becomes imperative for responsive and data-driven urban planning, ensuring that transportation systems align seamlessly with the dynamic needs of urban populations.

Several taxi trips observations where the distance trips between 13 km and 15 km are listed out using the given SAS query. After the query has been executed, one of the outputs of pickup location is selected and a map calculator is used to prove whether the given latitude and longitude are coordinates around the Bronx. As the result, the map output shows the coordinates are pointed at The Bronx area which is about 13 km away from the city center.

Additionally, several taxi trips observations where the distance trips greater than 20 km are listed out using the given SAS query. After the query has been executed, one of the outputs of pickup location is selected and a map calculator is used to prove whether the given latitude and longitude are coordinates around either Queens or JFK Airport area. As the result, the output of the map shows the coordinates are pointed at Queens area which is about 20 km away from the city center.

IV. RESULTS AND DISCUSSION

To sum up, data is the biggest asset for urban city planning. Geographic information helps the government or institution to have a better understanding of urban city in order to solve the real-world problem such as optimizing transportation resources, urban mobility and many more. However, this insight is hardly obtained without data exploration process. Therefore, data preparation and feature engineering are the necessary tasks to organizing major events, reduce traffic congestion, sensing the transform raw information into

meaningful insight of the traffic condition of big cities like New York. With increasing amounts of urban data being collected and made available, new possibilities of data-driven research emerge lead to changes in traffic condition through scientific proof-based decision-making and strategies. Comparison of prediction accuracy and error rate [16] with existing methods are given in table I.

TABLE I. COMPARISON OF PREDICTION ACCURACY AND ERROR RATE [16] WITH EXISTING METHODS

Methods	Accuracy	MAE
Mahmoud, H et al. [14]	89.4	0.402
Winarno E et al. [15]	92.3	0.387
Proposed Method	94.6	0.361

V. CONCLUSION

The study focuses on analyzing shifts in mobility within New York City by leveraging crowd-sourced traffic congestion data. These data, derived from the collective input of individuals navigating the city, offer valuable insights that extend beyond mere documentation of current mobility patterns. Their utility lies in their potential to inform real-time human mobility changes, strategically plan for future responses, and gain a comprehensive understanding of how large-scale traffic interventions, such as congestion pricing regulations, might impact urban dynamics. Crowd-sourced traffic congestion data serve as a dynamic lens through which the study gains a granular view of the current state of human mobility. This real-time approach, drawing from the experiences of those navigating the city's intricate traffic patterns, offers a more responsive and nuanced understanding compared to traditional data collection methods. It provides a valuable resource for deciphering the challenges and intricacies inherent in the urban transportation ecosystem. The study recognizes the proactive role that crowd-sourced data can play in informing decision-making processes related to human mobility. Beyond offering a retrospective snapshot, these data become a tool for anticipating and planning for future contingencies. By discerning patterns, anomalies, and emerging trends within the data, city planners and policymakers can adapt strategies to meet the evolving needs of a dynamic urban environment, ensuring a more agile and responsive approach to urban mobility challenges. Additionally, the study acknowledges the significance of crowd-sourced data in comprehending the potential impact of large-scale traffic interventions, specifically congestion pricing regulations. As cities grapple with the implementation of such measures, understanding how they may ripple through the urban transportation network becomes crucial. The study aims to unravel these potential impacts, shedding light on how congestion pricing regulations might influence commuter behavior, route choices, and overall traffic dynamics.

The predictive aspect of the envisioned research extension introduces a strategic dimension to urban planning. By harnessing insights from crowd-sourced data, the research aims to develop models capable of anticipating mobility patterns. This foresight becomes invaluable for city planners and policymakers seeking to address the challenges posed by growing urban populations and evolving societal dynamics. Predicting taxi flow and public transportation usage across diverse contexts positions this research at the forefront of shaping the future of urban mobility. In conclusion, the study showcases the transformative potential of crowd-sourced traffic congestion data in understanding and shaping urban

mobility. Beyond their descriptive capabilities, these data offer a real-time tool for informing current mobility changes, planning for future scenarios, and comprehending the potential impact of large-scale traffic interventions. The envisioned extension of the research to predict mobility patterns across different cities and time periods marks a strategic leap toward influencing the future of urban transportation. By embracing the predictive dimension and considering the diverse dynamics of urban landscapes, this research aspires to be a pioneering force in the ongoing discourse on sustainable and efficient urban mobility systems.

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